BUSS386 Problem Set 6 — Solutions

Introduction to Options

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Problem 1 — Terminology check

(a) A forward binds *both* sides to trade at T. An option gives the **buyer** the right to walk away; only the **seller** is bound.

(b) Intrinsic value = $\max(S_t - K, 0)$ (call) or $\max(K - S_t, 0)$ (put); always ≥ 0. Time value = price − intrinsic; also ≥ 0 but $\to 0$ as $t \to T$ and $\boxed{= 0 \text{ at expiry}}$.

(c) Short put = collects premium today, loses if the underlying becomes more volatile (a wider distribution increases the chance of a deep ITM put). She is therefore $\boxed{\text{short volatility}}$.

(d) *Volume* = contracts traded today (flow). *Open interest* = contracts currently outstanding (stock).

(e) Only $\boxed{\text{the Apple holder}}$ can exercise today. US listed single-stock options are **American** — exercisable on any business day up to expiry — and **physically settled**, so each contract exercised delivers 100 shares at the strike. The KOSPI 200 call is **European** and **cash-settled**: it can be exercised only on the expiry date (the second Thursday of the contract month), and it pays the cash difference × ₩250,000 per index point. Every KRX listed option is European and cash-settled. Note the two properties are independent: OEX, on the S&P 100, is American *and* cash-settled.

Problem 2 — Payoff and profit table

(a) Long call payoff = $\max(S_T - 100, 0)$, profit = payoff −6. Short call mirror. Long put payoff = $\max(100 - S_T, 0)$, profit = payoff −4. Short put mirror.

	Long call		Short call		Long put		Short put	
S_T	payoff	profit	payoff	profit	payoff	profit	payoff	profit
80	0	−6	0	+6	20	+16	−20	−16
95	0	−6	0	+6	5	+1	−5	−1
100	0	−6	0	+6	0	−4	0	+4
110	10	+4	−10	−4	0	−4	0	+4
120	20	+14	−20	−14	0	−4	0	+4

(b) Breakeven long call: $S_T = K + c = \boxed{\$106}$. Breakeven long put: $S_T = K - p = \boxed{\$96}$.

(c) $\boxed{\text{Short call}}$ has the larger max loss — unbounded as $S_T \to \infty$. Short put's max loss is bounded by $K - 0 - p = \$96$ (stock can't go below zero).

Problem 3 — KOSPI 200 option example

Multiplier $M = ₩\,250{,}000$ per index point; 5 contracts.

(a) Spot 1,105 < strike 1,120 ⇒ call is $\boxed{\text{OTM}}$.

(b) Intrinsic = $\max(1{,}105 - 1{,}120,\ 0) = \boxed{0}$. Time value = price − intrinsic = $26.60 - 0 = \boxed{26.60}$ pts. An OTM option is *all* time value.

(c) Premium = $5 \times 26.60 \times 250{,}000 = \boxed{₩\,33{,}250{,}000}$.

(d) Total profit at expiry = $5 \times M \times [\max(S_T - 1{,}120,\ 0) - 26.60]$:

S_T	Payoff/contract (pts)	Total profit (₩)
1,080	0	−33,250,000
1,120	0	−33,250,000
1,146.60	26.60	0
1,180	60	+41,750,000
1,220	100	+91,750,000

(e) Breakeven at $S_T = K + c = 1{,}120 + 26.60 = \boxed{1{,}146.60}$.

(f) Required index move: $(1{,}146.60 - 1{,}105)/1{,}105 = \boxed{3.76\%}$. A direct index investment profits the moment the index ticks up; the call must first climb 3.76% simply to recover its premium. Past that point the comparison reverses: the call gains ₩ 1,250,000 per index point, while the same ₩ 33.25M placed directly in the index gains only ₩ 30,090 per point — about $\boxed{41\times}$ the sensitivity. That asymmetry is the leverage trade-off.

Problem 4 — Insurance with a long put

(a) 200 shares ÷ 100 shares/contract = $\boxed{2}$ puts. Premium = $2 \times 100 \times \$8 = \boxed{\$1{,}600}$.

(b) Stock value = $200 \cdot S_T$. Put payoff = $200 \cdot \max(475 - S_T, 0)$. Hedged value = stock + put payoff − premium:

S_T	Stock (×200)	Put payoff	Hedged value (net of \$1,600)
400	80,000	15,000	$80{,}000 + 15{,}000 - 1{,}600 = \boxed{\$93{,}400}$
450	90,000	5,000	$90{,}000 + 5{,}000 - 1{,}600 = \$93{,}400$
475	95,000	0	$95{,}000 - 1{,}600 = \$93{,}400$
500	100,000	0	$100{,}000 - 1{,}600 = \$98{,}400$
550	110,000	0	$110{,}000 - 1{,}600 = \$108{,}400$

(c) Worst-case = floor at $\boxed{\$93{,}400}$ = $K \cdot 200$− premium = $475 \cdot 200 - 1{,}600$, achieved for any $S_T \le 475$.

(d) Selling outright eliminates upside (gives up the $S_T = 550$ gain). Selling futures locks in a single price (no choice). The put preserves upside above \$475 at the cost of the \$1,600 premium (the "insurance deductible").

Problem 5 — Where is the option?

(a) **Convertible bond** = straight bond + long call on the issuer's stock. Conversion strike = \$1,000/25 = \$40 per share. The issuer is short the call, and exercise is *dilutive* — the shares are newly issued, not bought in the market. The holder accepts 1% instead of 5% because the four percentage points of forgone yield *are* the option premium, paid in instalments over the life of the bond rather than up front.

(b) **Employee stock option.** Long a call on your employer's stock, strike ₩ 60,000, written by the company; existing shareholders are diluted on exercise. It is worth less than 2,000 identical exchange-traded calls for three reasons: (i) it is **non-transferable**, so the only way to turn it into cash is to exercise, which throws away the remaining time value; (ii) it is **forfeited** if you leave before vesting, so it is really a call conditional on continued employment; and (iii) you are **undiversified** — your salary is already a claim on the same firm — so you value the payoff below what a diversified investor would.

(c) **Deposit insurance is a put.** The underlying is the *bank's assets* (equivalently, your claim on the bank). You are long a put with strike equal to your balance, capped at ₩ 50m: if the bank fails you hand over a worthless claim and receive the strike in cash. The insurer — KDIC, and ultimately the taxpayer — is short it, and receives a premium from the bank. This is Merton's (1977) observation; the same logic makes equity in a levered firm a call on the firm's assets.

(d) **A real option** — a call on the sequel project's NPV, with strike equal to the agreed production budget. The important difference: nobody is contractually short. The option arises from *owning the asset*, not from a counterparty's promise. So there is no market price to read off a screen, no counterparty risk, and valuing it means modelling the project rather than quoting it.